CHAPTER 7:

Currency Futures and Options Markets

Chapter Outline

- Overview
- Futures market
- Currency options market
- Option strategies
- Comparing forward or futures contracts with options contracts

Overview

- What are futures and options contracts?
- A brief history of currency derivatives

What are Futures and Options Contracts?

- Futures and options are **derivative securities**: Their value is determined or derived by the value of some underlying asset (in this case, a currency pair).
- Other types of derivatives
  - Forward contracts
  - Swaps (A swap rate is a fixed interest rate that's agreed upon by two parties in a swap agreement. Swaps can be used for interest rates, currencies, or commodities

Derivatives traders

Hedgers

Goal: reduce or eliminate their risk exposure

- E.g., airlines want to reduce their exposure to increases in oil prices)

Speculators

Goal: profit from taking on the risk

Arbitrageurs

Goal: make riskless profits with little or no invested capital by exploiting deviations between the derivative's price and the price implied by the underlying asset

What is a Futures Contract?

- An agreement between 2 parties to buy or sell a specific ***quantity*** of a specific ***asset*** at a specific ***price*** at a specific ***time*** in the future.

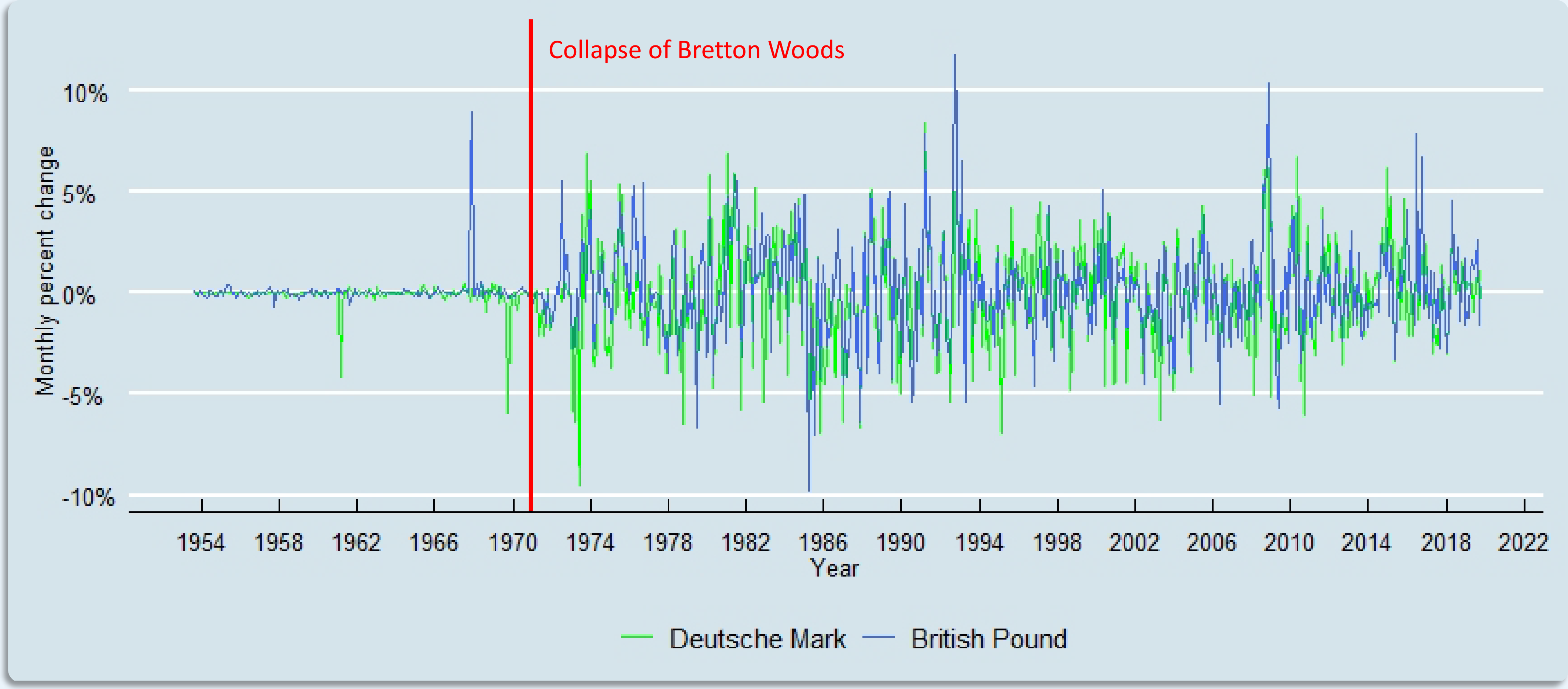
Specific quantity →	Contract size
Specific asset →	Reference, or underlying, asset (e.g., wheat, oil, currencies)
Specific price →	Futures price
Specific time →	Expiration date (or settlement or maturity date)

A Brief History of Currency Derivatives

- While currency derivatives have existed for a long time, their popularity soared in the 1970s.
- In 1971, the Bretton Woods system of fixed exchange rates ended and exchange rates became highly volatile.
- Under the fixed exchange-rate system, MNCs bore little foreign exchange-rate risk and found little use for currency derivatives.
- After the collapse of Bretton Woods, MNCs faced significant exchange-rate risk exposures to their domestic and foreign operations.
- Changes in exchange rates could make foreign competitors more (or less) attractive (both at home and abroad), increase (or decrease) the MNCs costs from foreign suppliers, decrease (or increase) the value of foreign assets, etc.

A Brief History of Currency Derivatives

Currency volatility post-Bretton Woods



Futures Market

- What is a futures contract?
- Differences between forwards and futures
- Currency futures exchanges
- CME Group futures trading
- Margins and marking-to-market
- Trading futures: example
- How MNCs use futures to hedge

What is a Futures Contract?

- An agreement between 2 parties to buy or sell a specific ***quantity*** of a specific ***asset*** at a specific ***price*** at a specific ***time*** in the future.

Specific quantity	→	Contract size
Specific asset	→	Reference, or underlying, asset (e.g., wheat, oil, currencies)
Specific price	→	Futures price
Specific time	→	Expiration date (or settlement or maturity date)

Differences between Forwards and Futures

Forwards	Futures
Over the counter ("OTC")	Exchange-traded
Self-regulating	Regulated by the Commodity and Futures Trading Commission
> 90% settled by actual delivery	< 1% have actual delivery
Customized contract size	Standardized contract size
Any delivery date	Delivered on specific dates a year
Quoted (sometimes) in units of foreign currency per $. However, students do not need to be too concerned about this issue and simply follow the examples in this PPT slides and tutorials)	Quoted (sometimes) in $ for one unit of foreign currency
Transaction costs are based on bid-ask spread	Transaction costs based on broker fees
Margins not required	Margins are required
Counterparty risk exists for both sides	The exchange assumes counterparty risk

Currency Futures Exchanges

- Globally, 19 exchanges offer currency futures.
- Major exchanges
  - The Chicago Mercantile Exchange (CME) Group (largest exchange measured by notional value)
  - ICE Futures Europe
  - Eurex based in Frankfurt and Zurich
- Smaller independent exchanges in Singapore, Sao Paolo, Bombay, Moscow, Seoul, Johannesburg, …).
- The Australian Securities Exchange (ASX) trades in futures contracts through ASX 24. ASX 24 is a market for trading futures and options on commodities, equity, and interest rates

CME Group Futures Trading

- The world's leading derivatives marketplace.
- Its Globex electronic platform trades currency futures 23 hours a day, Sunday to Friday.
- Offers futures on a wide range of currency pairs and occasionally introduces (or discontinues) new (existing) currency pairs.
- Also offers smaller contracts. E.g.:
  - Regular EUR/USD future is €125,000.
  - E-mini is €62,500,
  - E-micro is €12,500.

For the next Slide, please note the following definitions

- “Last” is the current price; i.e., the most recent trade at the time given in the column, “Updated”.
- “Prior Settle” is the official closing price from the previous trading day.
- “Open” is the first traded price of the current trading session.
- “Change” is the difference between the current price, “Last”, and the prior settle.

CME Group Futures Trading

British Pound Futures Quotes

Globex

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Globex Futures | Globex Options | Open Outcry Options

Auto Refresh Is ON

Market data is delayed by at least 10 minutes.

All market data contained within the CME Group website should be considered as a reference only and should not be used as validation against, nor as a complement to, real-time market data feeds. Settlement prices on instruments without open interest or volume are provided for web users only and are not published on Market Data Platform (MDP). These prices are not based on market activity.

Month	Options	Charts	Last	Change	Prior Settle	Open	High	Low	Volume	Hi / Low Limit	Updated
NOV 2019	OPT		**1.2830**	-0.0046	1.2876	1.2845	1.2864	1.2815	428	1.3276 / 1.2476	11:35:07 CT 25 Oct 2019
DEC 2019	OPT		**1.2840**	-0.0047	1.2887	1.2866	1.2884	1.2825	69,849	1.3287 / 1.2487	11:55:01 CT 25 Oct 2019
JAN 2020	OPT		-	-	1.2903	-	-	-	1,809	1.3303 / 1.2503	08:33:34 CT 25 Oct 2019
FEB 2020	OPT		**1.2878**	-0.0036	1.2914	1.2878	1.2878	1.2878	1	1.3314 / 1.2514	08:19:35 CT 25 Oct 2019
MAR 2020	OPT		**1.2880**	-0.0044	1.2924	1.2900	1.2900	1.2880	12	1.3324 / 1.2524	11:27:19 CT 25 Oct 2019
JUN 2020	OPT		**1.2888**	-0.0064	1.2952	1.2888	1.2888	1.2888	1	1.3352 / 1.2552	08:49:57 CT 25 Oct 2019

Contract size	\$62,500
Underlying asset	Great Britain pounds (GBP)
Future price	\$1.2888
Expiration date	15-Jun-2020 (at 9:16am CT, on the second business day immediately preceding the third Wednesday of the contract month).

Margins and Marking-to-Market

Initial margin

- **“Initial performance bond”***
- Cash that must be deposited when contract is entered into.
- Ensures sufficient funds to prevent default.

Maintenance margin

- **“Maintenance performance bond”***
- Minimum cash balance required in account.
- If cash < maintenance, triggers **margin call** to bring account to initial level.
- If cash > initial level, investor can withdraw the excess.

Marking-to-market

- Profits and losses are credited to or debited from investor’s account at end of each trading day (**daily settlement**).
- Settlement occurs at 2pm CT for currency futures on the CME Globex.

Margins depend on the risk of a particular contract and are updated as the risks change, calculated by the CME Group’s computerized risk management program, *SPAN* (Standard Portfolio Analysis of Risk).

*For the CME Group

Margins and Marking-to-Market

- On Monday morning at 9am CT, you go long on the CME euro futures contract:
  - Future price: €1 = $1.10
  - Contract size: EUR 125,000.
  - Initial margin: $2,530
  - Maintenance margin: $2,000

Day	Future Price	Profit/ Loss	Margin Balance	Margin Call
Monday (9am CT)	$1.100	$0	$2,530	
Monday (2pm CT)	$1.105	$625	$3,155	
Tuesday (2pm CT)	$1.100	-$625	$2,530	
Wednesday (2pm CT)	$1.097	-$375	$2,155	
Thursday (2pm CT)	$1.095	-$250	$1,905	$625
Friday (2pm CT)	$1.095	$0	$2,530	

Let’s look at the daily **profit/loss** in more detail.

- Monday 9am, price = \$1.100
- Monday 2pm, price = \$1.105.
- Change in price = + 0.005 (50 pips).
- Contract size 125,000 euros.
- Profit = $+0.005 \times 125{,}000$ = \$625.
- Tuesday 2pm, price = \$1.100 (lose \$625)
- Wednesday 2pm, price = \$1.097 (lose \$$0.003 \times 125{,}000$ = \$375)
- Thursday 2pm, price = \$1.095 (lose \$$0.002 \times 125{,}000$ = \$250)
- Margin call: \$1,905 + \$625 = **\$2,530**. (NOT \$2,000)
- Friday: no change.

Margins and Marking-to-Market

Day	Action	Cash Flow
Tuesday (9am CT)	An investor buys a CHF (Swiss francs) futures contract that matures in two days at agreed on price rate \$0.95 (size of contract is CHF125,000). Please see Table 7.1 of the textbook regarding CHF information.	None
Tuesday (2pm CT)	▪ Futures prices rises to \$0.955 ▪ Position is marked-to-market	▪ 125,000 x (0.955 – 0.95) = \$625 gain ▪ Investor **receives \$625**
Wednesday (2pm CT)	▪ Futures price drops to \$0.943 ▪ Position is marked-to-market	▪ 125,000 x (0.943 – 0.955) = -\$1,500 loss ▪ Investor **pays \$1,500**
Thursday (2pm CT)	▪ Futures price drops to \$0.94 ▪ Contract is marked-to-market	▪ 125,000 x (0.94 – 0.943) = -\$375 loss ▪ Investor **pays \$375**
	▪ Investor takes delivery of CHF125,000	125,000 x 0.94 = \$117,500 \$117,500 – (125,000 x 0.95) = **-\$1,250 loss on contract**

- **Read on your own:** Let's consider the same contract, but with a *short* position. In this case, we make money if the futures price *decreases*.

Day	Futures Price	Profit/ Loss	Margin Balance	Margin Call
Monday (9am)	$1.100	$0	$2,530	
Monday (2pm)	$1.105	–$625	$1,905	$625
Tuesday (2pm)	$1.100	+$625	$3,155	
Wed. (2pm)	$1.097	+$375	$3,530	
Thurs. (2pm)	$1.095	+$250	$3,780	
Friday (2pm)	$1.095	$0	$3,780	

Trading Futures: Example

- An investor agrees to buy £62,500 on June 15, 2020, at $1.2888/£.
- The futures price changes during the trading day throughout the life of the contract.
  - I.e., another buyer might buy a contract with the same expiration date at $1.290/£.
  - This new buyer would promise to buy £62,500 on June 15, 2020 at $1.290/£.
- The buyer pays nothing today. Buyer’s options.

Sell prior to expiration

Sell the contract prior to expiration
(99% of all cases)

Wait until expiration

Make delivery of
$1.2888 x 62,500 = $80,550
(1% of all cases)

Close prior to expiration

Make an **offsetting trade:** Sell the identical contract prior to expiration. The two positions cancel out on the books.

Trading Futures: Example

- Assume the futures price on the same contract is now $1.300/£.
  - The buyer can offer to sell a futures contract that agrees to pay $1.300/£ on June 15, 2020.
  - The buyer now has two positions (one is long at $1.288/£ and the other is short at $1.300/£).

On June 15, 2020, the buyer pays on the first contract:	But receives from the second contract:	And locks in a profit of:
$1.288/£ x £62,500 = $80,500	$1.300/£ x £62,500 = $81,250	$81,250 - $80,500 = $750

  - When the position is offset, the exchange automatically cancels out the two trades, and the profit is registered in the account.

How MNCs Use Futures to Hedge

- Chicago-based Boeing agrees to purchase equipment in four months from Mecadaq Group, a French supplier, for its 787 Dreamliner.
- The equipment costs €2,500,000, payable in four months.

  - The current spot price for the Euro is: S_0= \$1.20/€
  - 4-month futures price: F= \$1.15/€
  - Contract size = €125,000.

- Boeing faces exchange rate risk and would like to hedge.
- What is the position (and size) of the futures hedge? Buy 20 futures contracts.

How MNCs Use Futures to Hedge

What happens in 4 months to Boeing's hedged cost for the equipment, if…

➡ The euro **drops** to $1.10/€?

Spot market cost:

2,500,000 x $1.10 = $2,750,000

Futures market transaction:

€2,500,000 x ($1.10 – $1.15) = -$125,000 (loss)

Total cost to Boeing:

$2,750,000 + $125,000 = $2,875,000

Implied rate:

$2,875,000 / $2,7500,000 = $1.15/€

➡ The euro **increases** to $1.30/€?

Spot market cost:

2,500,000 x $1.30 = $3,250,000

Futures market transaction:

€2,500,000 x ($1.30 – $1.15) = $375,000 (gain)

Total cost to Boeing:

$3,250,000 – $375,000 = $2,875,000

Implied rate:

$2,875,000 / $2,7500,000 = $1.15/€

No matter what happens to the price of the €, Boeing has locked in a cost of $1.15/€.

Currency Options Market

- What is an options contract?
- Profit and loss from buying and selling calls and puts
- Examples of an MNC using calls and puts

What is an Options Contract?

- A financial instrument giving the holder (buyer) the **right,** but not the **obligation,** to buy or sell a ***specific quantity*** of a ***specific asset*** at a ***specific price*** up to a ***specific date***.

Contract Characteristic	Referred to as
Quantity	Contract size
Asset	Underlying asset
Price	Strike, or exercise, price
Date	Expiration date

- The seller ("writer") of the put or call option must fulfill the contract if the buyer so desires.
- Because the option not to buy or sell has value, the buyer must pay the seller a **premium**.

What is an Options Contract?

Types of options

Call option

Gives the holder the right to **buy** at a specific price

Put option

Gives the holder the right to **sell** at a specific price

American option

Can exercise any time **up to and including** the expiration date

European option

Can exercise **only on** the expiration date

Notation

K = strike price

S = spot rate

What is an Option Contract?

- Options are ***in-the-money, at-the-money*** or ***out-of-the-money,*** depending on the strike price relative to the spot rate (i.e., K relative to S).

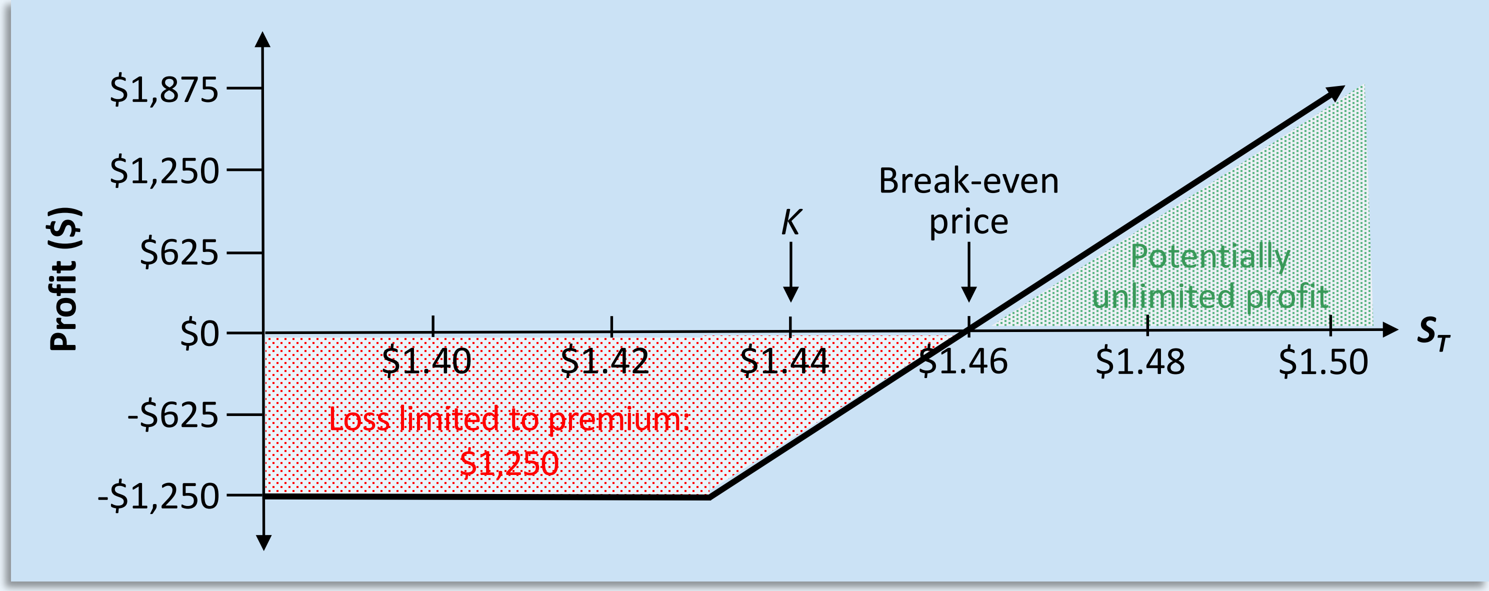
  - The **call** buyer wants S to ***increase above*** K in order to ***buy*** the underlying currency at a profit (i.e., to exercise) or see the value of the option increase (i.e., to sell).
  - The **put** buyer wants S to ***decrease below*** K in order to ***sell*** the underlying currency at a profit (i.e., to exercise) or see the value of the option increase (i.e., to sell).

Option Description	Strike Price Relative to Spot Rate		Effect of Exercise
	Call (buy) Option	Put (sell) Option	
In-the-money	Strike < Spot ($K < S$)	Strike > Spot ($K > S$)	Profit
At-the-money	Strike = Spot ($K = S$)	Strike = Spot ($K = S$)	Indifference
Out-of-the-money	Strike > Spot ($K > S$)	Strike < Spot ($K < S$)	Loss

Profit and Loss from Buying and Selling Calls and Puts

Buying a call

- Contract size = €62,500
  - K = $1.44/€
  - Option premium = $0.02, or $1,250/contract (62,500 x 0.02 = $1,250)
  - Expiration date is 60 days
  - If exercise the option, pay 62,500 x 1.44 = $90,000.
  - Option payoff if exercise: 62,500 x (S_T - K), where S_T > K
  - Otherwise, payoff = 0.

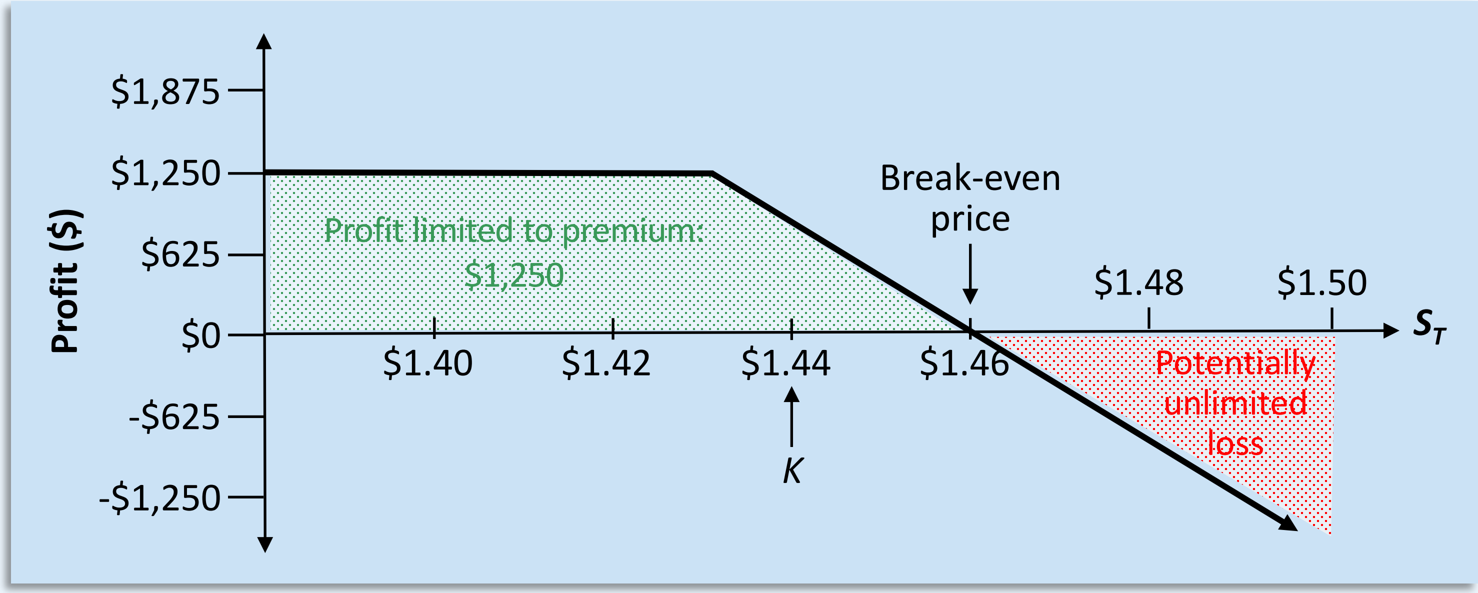


	Out-of-the-money		At-the-money	In-the-money		
	$1.40	**$1.42**	**$1.44**	**$1.46**	**$1.48**	**$1.50**
Inflows						
Spot sale of euros	--	--	--	91,250	92,500	93,750
Outflows						
Call premium	-1,250	-1,250	-1,250	-1,250	-1,250	-1,250
Exercise of option	--	--	--	-90,000	-90,000	-90,000
Profit (loss)	-$1,250	-$1,205	-$1,250	$0	$1,250	$2,500

Profit and Loss from Buying and Selling Calls and Puts

Selling a call

- Contract size = €62,500
  - *K* = $1.44/€
  - Option premium = $0.02, or $1,250/contract (62,500 x 0.02 = $1,250)
  - Expiration date is 60 days

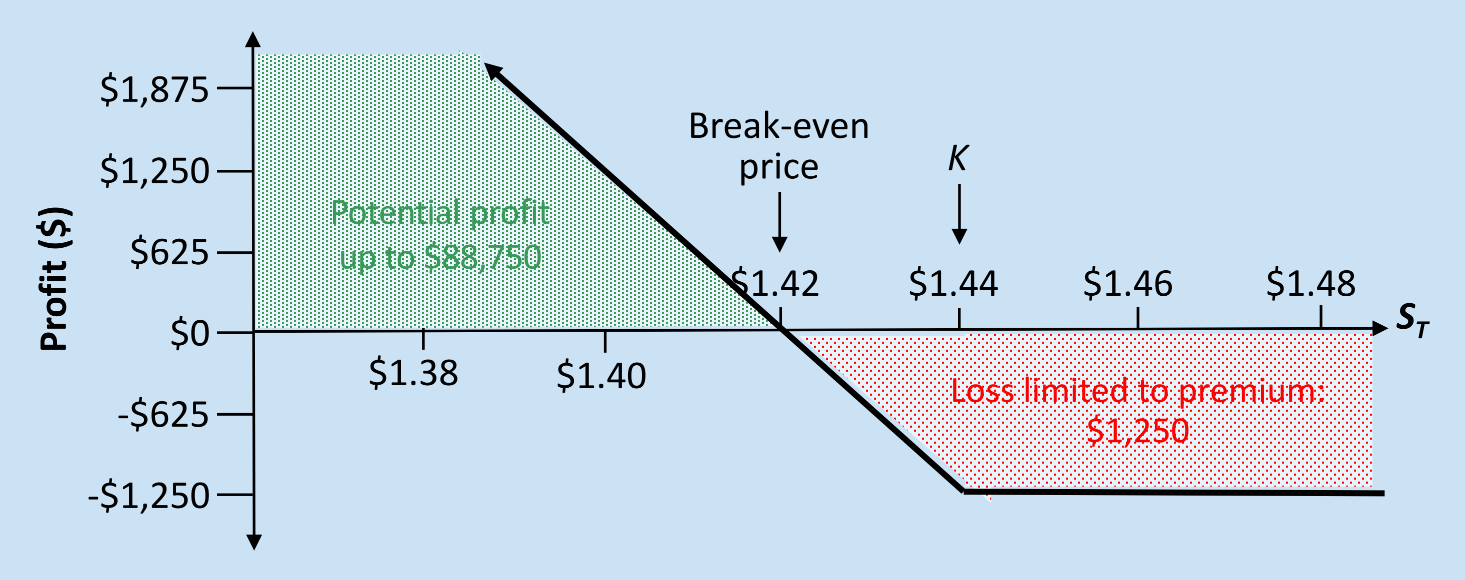


	Out-of-the-money		At-the-money		In-the-money	
Inflows	**$1.40**	**$1.42**	**$1.44**	**$1.46**	**$1.48**	**$1.50**
Call premium	1,250	1,250	1,250	1,250	1,250	1,250
Exercise of option	--	--	--	90,000	90,000	90,000
Outflows						
Spot purchase of euros	--	--	--	-91,500	-92,500	-93,750
Profit (loss)	$1,250	$1,250	$1,250	$0	-$1,250	-$2,500

Profit and Loss from Buying and Selling Calls and Puts

Buying a put

- Contract size = €62,500
  - K = \$1.44/€
  - Option premium = \$0.02, or \$1,250/contract (62,500 x 0.02 = \$1,250)
  - Expiration date is 60 days
  - Option payoff if exercise: 62,500 x (K - S_T), where S_T < K.
  - Otherwise, payoff = 0.

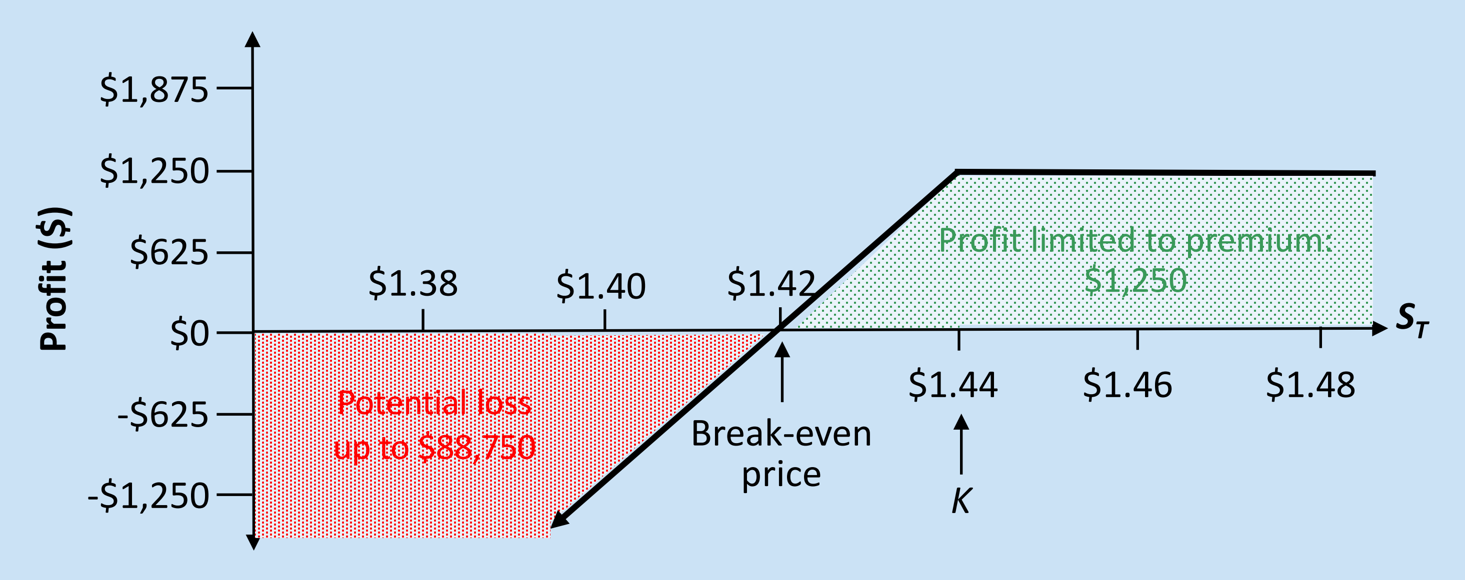


	In-the-money			At-the-money	Out-of-the-money	
Inflows	**\$1.38**	**\$1.40**	**\$1.42**	**\$1.44**	**\$1.46**	**\$1.48**
Exercise of option	90,000	90,000	90,000	--	--	--
Outflows						
Put premium	-1,250	-1,250	-1,250	-1,250	-1,250	-1,250
Spot purchase of euros	-86,250	-87,500	-88,750	--	--	--
Profit (loss)	\$2,500	\$1,250	\$0	-\$1,250	-\$1,250	-\$1,250

Profit and Loss from Buying and Selling Calls and Puts

Selling a put

- Contract size = €62,500
  - K = \$1.44/€
  - Option premium = \$0.02, or \$1,250/contract (62,500 x 0.02 = \$1,250)
  - Expiration date is 60 days



	In-the-money			At-the-money	Out-of-the-money	
Inflows	**\$1.38**	**\$1.40**	**\$1.42**	**\$1.44**	**\$1.46**	**\$1.48**
Put premium	1,250	1,250	1,250	1,250	1,250	1,250
Spot sale of euros	86,250	87,500	88,750	--	--	--
Outflows						
Exercise of option	-90,000	-90,000	-90,000	--	--	--
Profit (loss)	-\$2,500	-\$1,250	\$0	\$1,250	\$1,250	\$1,250